Assignment #3: 20260311 cs201 Mock Exam

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1. 题目

E20742:泰波拿契數

implementation, http://cs101.openjudge.cn/practice/20742/

思路:

代码:

```
n = int(input())
def dfs(n):
    if n == 0:
        return 0
    if n == 1:
        return 1
    if n == 2:
        return 1
    return dfs(n-1)+dfs(n-2)+dfs(n-3)
print(dfs(n))
```

alt text

E30571.十进制整数的反码

bit manipulation, http://cs101.openjudge.cn/practice/E30571/

思路:

代码:

```python
import math
n = int(input())
t = int(math.log2(n))+1
print((1<<t)-1-n)
```

#51937750提交状态

View Submit Statistics Clarify

状态: Accepted

Source Code

```python
import math
n = int(input())
t = int(math.log2(n))+1
print((1<<t)-1-n)
```

基本信息

#: 51937750
题目: E30571
提交人: 25n2500010774(sleepy(25n2500010774)
内存: 3604kB
时间: 26ms
语言: Python3
提交时间: 2026-03-11 15:20:02

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中文 Help About

E29950:稳定的符文序列

two pointers, http://cs101.openjudge.cn/practice/E29950

M30218:狭路相逢

stack, http://cs101.openjudge.cn/practice/M30218/

思路:

代码:

```python
n = int(input())
mylist = list(map(int,input().split()))
result = []
for num in mylist:
    if not result:
        result.append(num)
    else:
        flag = True
        while result:
            a = result[-1]
            if num < 0:
                if a < 0:
                    result.append(num)
                    flag = False
                    break
                else:
                    if a > -num:
                        result.pop()
                        result.append(a+num)
                        flag = False
                        break
                    else:
                        result.pop()
                        num += a
            elif num > 0:
                result.append(num)
                flag = False
                break
            elif num == 0:
                flag = False
                break
        if flag:
            result.append(num)
print(len(result))
print(*result)
```

#51938954提交状态

View Submit Statistics Clarify

状态: Accepted

Source Code

```
n = int(input())
mylist = list(map(int,input().split()))
result = []
for num in mylist:
    if not result:
        result.append(num)
    else:
        flag = True
        while result:
            a = result[-1]
            if num < 0:
                if a < 0:
                    result.append(num)
                    flag = False
                    break
                else:
                    if a > -num:
                        result.pop()
                        result.append(a+num)
```

基本信息

#: 51938954
题目: M30218
提交人: 25n2500010774(sleepy(25n2500010774)
内存: 3676kB
时间: 26ms
语言: Python3
提交时间: 2026-03-11 16:08:03

M02299: Ultra-QuickSort

merge sort, http://cs101.openjudge.cn/practice/02299/

思路:

代码:

```python
def ultrasort(mylist):
    count = 0
    def mergesort(i,j):
        nonlocal count
        if i == j:
            return [mylist[i]]
        med = (i+j)//2
        list1 = mergesort(i,med)
        list2 = mergesort(med+1,j)
        t,s = -1,-1
        list3 = []
        while True:
            if t == len(list1)-1:
                list3.extend(list2[s+1:])
                break
            if s == len(list2)-1:
                list3.extend(list1[t+1:])
                break
            if list1[t+1] < list2[s+1]:
                list3.append(list1[t+1])
                t += 1
            else:
                list3.append(list2[s+1])
                s += 1
                count += len(list1)-t-1
        return list3
    mergesort(0,len(mylist)-1)
    return count
while True:
    n = int(input())
    if n == 0:
        break
    mylist = []
    for _ in range(n):
        mylist.append(int(input()))
    print(ultrasort(mylist))
```

#51938106提交状态

View Submit Statistics Clarify

状态: Accepted

Source Code

```
def ultrasort(mylist):
    count = 0
    def mergesort(i,j):
        nonlocal count
        if i == j:
            return [mylist[i]]
        med = (i+j)//2
        list1 = mergesort(i,med)
        list2 = mergesort(med+1,j)
        t,s = -1,-1
        list3 = []
        while True:
            if t == len(list1)-1:
                list3.extend(list2[s+1:])
                break
            if s == len(list2)-1:
                list3.extend(list1[t+1:])
                break
            if list1[t+1] < list2[s+1]:
```

基本信息

#: 51938106
题目: M02299
提交人: 25n2500010774(sleepy(25n2500010774)
内存: 34112kB
时间: 4917ms
语言: Python3
提交时间: 2026-03-11 15:36:16

M29954:逃离紫罗兰监狱

bfs, http://cs101.openjudge.cn/practice/29954

思路:

代码:

```
r,c,k = map(int,input().split())
matrix = []
for _ in range(r):
    mylist = list(input())
    matrix.append(mylist)
    if "S" in mylist:
        a = mylist.index("S")
        sx,sy = _,a
    if "E" in mylist:
        a = mylist.index("E")
        ex,ey = _,a
from collections import deque
def bfs():
    queue = deque([(sx,sy,0,0)])
    visited = {(sx,sy,0)}
    vectors = [(0,1),(1,0),(-1,0),(0,-1)]
    while queue:
        a,b,step,mag = queue.popleft()
        for i,j in vectors:
            if a+i == ex and b+j == ey:
                return step+1
            if a+i >= 0 and a+i <= r-1 and b+j >= 0 and b+j <= c-1:
                if matrix[a+i][b+j] == "." and (a+i,b+j,mag) not in visited:
                    queue.append((a+i,b+j,step+1,mag))
                    visited.add((a+i,b+j,mag))
                if matrix[a+i][b+j] == "#" and (a+i,b+j,mag+1) not in visited and mag <
                    queue.append((a+i,b+j,step+1,mag+1))
                    visited.add((a+i,b+j,mag+1))
    return -1
print(bfs())
```

#51938479提交状态

View Submit Statistics Clarify

状态: Accepted

Source Code

```
r,c,k = map(int,input().split())
matrix = []
for _ in range(r):
    mylist = list(input())
    matrix.append(mylist)
    if "S" in mylist:
        a = mylist.index("S")
        sx,sy = _,a
    if "E" in mylist:
        a = mylist.index("E")
        ex,ey = _,a
from collections import deque
def bfs():
    queue = deque([(sx,sy,0,0)])
    visited = {(sx,sy,0)}
    vectors = [(0,1),(1,0),(-1,0),(0,-1)]
    while queue:
        a,b,step,mag = queue.popleft()
        for i,j in vectors:
```

基本信息

#: 51938479
题目: M29954
提交人: 25n2500010774(sleepy(25n2500010774)
内存: 15044kB
时间: 311ms
语言: Python3
提交时间: 2026-03-11 15:51:29

2. 学习总结和个人收获

如果发现作业题目相对简单，有否寻找额外的练习题目，如“数算2025spring每日选做”、LeetCode、Codeforces、洛谷等网站上的题目。